

SMART SHOPPING TROLLEY

A MINI PROJECT REPORT

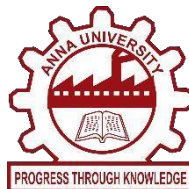
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In partial fulfillment of the requirements for the degree

Of

**BACHELOR OF ENGINEERING IN ELECTRICAL AND
ELECTRONICS ENGINEERING**



**ANNA UNIVERSITY REGIONAL CAMPUS COIMBATORE
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ABSTRACT

Conventional billing system used in the super market nowadays, are a big hassle for the customer as it creates long waiting time for the billing of the products purchased to provide a solution for the above problem we have designed an application called Smart Shopping Trolley using Arduino which can eliminate queues for billing throughout the supermarket. Using this trolley, we can scan the products and then put them inside the smart trolley which is designed using LCD display the total price of the items in the trolley. To make it more feasible the bill also send to the customer phone via message that makes it easy for the peoples to check the list of items added and so taken manage the purchase.

In the rapidly evolving landscape of retail, the demand for efficiency, customer convenience, and automation has grown significantly. One of the major pain points in traditional shopping is the time-consuming checkout process, where customers often have to stand in long queues to get their items billed. The **Smart Shopping Trolley** project aims to address this issue by integrating smart technologies into the conventional shopping cart to automate and streamline the shopping experience.

This project proposes the development of an intelligent shopping trolley using the **Arduino microcontroller platform**, equipped with **RFID (Radio Frequency Identification) technology** and a real-time **display system**. The fundamental concept revolves around allowing customers to scan items as they place them into the trolley, thereby eliminating the need for scanning items individually at the checkout counter.

Each product in the store is equipped with a unique RFID tag. The trolley is integrated with an **RFID reader** that scans these tags. When a customer places an item in the trolley, the RFID reader identifies the product and communicates with the Arduino, which processes the data. The product's name, price, and other relevant information are then displayed on an **LCD or OLED screen** attached to the trolley. The total bill is dynamically updated as more items are added or removed.

To ensure flexibility and connectivity, the system can be enhanced with **Bluetooth or Wi-Fi modules** such as the ESP8266 or HC-05. These modules allow the trolley to communicate with a central server, a billing counter, or even a mobile app, depending on the implementation. In this way, the final bill can

be transferred wirelessly, reducing the load at checkout counters and enabling a **self-checkout model**.

An optional **weight sensor** can be included to verify if the scanned product has been physically placed into the trolley, thereby minimizing the chances of theft or billing errors. A **buzzer** or **LED alert system** can also be incorporated to notify customers of invalid scans or unauthorized item removals.

The advantages of this smart system include significant **time savings**, **reduction in manpower** at billing counters, **improved inventory tracking**, and an overall **enhanced user experience**. From the retailer's perspective, it offers better customer engagement and the potential for **real-time data analytics** on shopping behavior.

The Arduino-based Smart Shopping Trolley is cost-effective, modular, and scalable. It demonstrates how embedded systems and IoT technologies can be leveraged to modernize retail infrastructure. Furthermore, the project promotes the idea of **smart retailing**, which aligns with the global trend of digital transformation in commercial spaces.

In conclusion, the Smart Shopping Trolley not only enhances the shopping experience by making it faster and more efficient but also serves as a stepping stone toward a fully automated and intelligent retail environment. The implementation of this system has the potential to transform the way we shop, making it smarter, simpler, and more customer-centric.

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CHAPTER-1

INTRODUCTION

1.1 OVERVIEW:

The Smart Shopping Trolley project is an innovative solution aimed at transforming the conventional retail shopping experience through automation and smart technology. It is designed to minimize the inconvenience faced by customers due to long queues at billing counters and to streamline the entire shopping and billing process. The project is centered around the use of the Arduino microcontroller, which serves as the core processing unit. The trolley is equipped with an RFID reader that scans the RFID tags attached to every product. As the customer places items into the trolley, the RFID reader reads each tag and sends the information to the Arduino. The system then processes the data and displays the item name and price on an LCD screen mounted on the trolley. The total bill is continuously updated in real-time as more items are added or removed.

This smart system enables customers to keep track of their purchases and the running total throughout their shopping journey. At the end of the shopping trip, the customer can proceed directly to a checkout kiosk or even pay wirelessly through an integrated app, thus bypassing traditional billing counters. The trolley can be further enhanced by integrating wireless modules such as Wi-Fi (ESP8266) or Bluetooth (HC-05), enabling data transfer to a central billing system or mobile device. Additional components such as buzzers and LEDs can be used for alerting the user in case of duplicate scans or unauthorized item removal. A weight sensor can be optionally included to cross-check the weight of scanned items, providing an extra layer of verification.

This project offers several advantages: it reduces manpower, enhances shopping efficiency, and contributes to digital retail transformation. It can be particularly useful in supermarkets, malls, and large retail outlets where customer traffic is high.

1.2 PROBLEM STATEMENT :

In traditional retail stores, customers often face long waiting times at billing counters due to the manual scanning of each item, leading to inconvenience, delays, and customer dissatisfaction. This process is inefficient, prone to human error, and requires significant manpower. Additionally, shoppers have no real-time visibility into their total expenditure while shopping, which can lead to confusion or overspending. To address these issues, there is a need for a smart, automated solution that can streamline the billing process, provide instant updates on purchased items, and enhance the overall shopping experience. The Smart Shopping Trolley using Arduino and RFID technology aims to eliminate these problems by enabling real-time item tracking and billing directly within the trolley, thereby reducing checkout time and improving operational efficiency in retail environments.

1.3 OBJECTIVE:

The main objective of the Smart Shopping Trolley project is to develop an automated system that enhances the shopping experience by integrating real-time item tracking and billing directly into the trolley using Arduino and RFID technology. This system aims to eliminate the need for manual item scanning at checkout counters, thereby reducing long queues, minimizing human error, and improving overall efficiency in retail environments. Additionally, it provides customers with instant feedback on their selected items and total cost through an onboard display, promoting transparency and convenience. The project also seeks to demonstrate how embedded systems and IoT can be effectively applied to modernize and digitize traditional retail operations.

1.4 SCOPE OF THE PROJECT:

The scope of the Smart Shopping Trolley project encompasses the design and development of a prototype system that integrates Arduino-based embedded technology with RFID and display modules to automate the shopping and billing process in retail stores. This project focuses on enabling real-time item identification, price calculation, and billing within the trolley itself, thus reducing reliance on traditional checkout counters. The system is scalable and can be implemented in supermarkets, grocery stores, and departmental outlets. It also opens up possibilities for future

enhancements such as mobile app integration, cloud-based inventory management, automated payment systems, and anti-theft mechanisms. By addressing both customer convenience and store operational efficiency, the project lays the groundwork for smarter and more seamless shopping experiences in the retail sector.

1.4 ADVANTAGES:

The Smart Shopping Trolley offers several significant advantages that enhance both the customer experience and store operations. By automating the billing process, it drastically reduces checkout time, allowing customers to avoid long queues and complete their purchases more quickly. The real-time billing feature enables shoppers to keep track of their total expenses as they add items to the trolley, promoting transparency and better budgeting. Automation minimizes human errors often encountered during manual billing, ensuring accurate price calculations. Additionally, the system reduces the need for extensive staff at checkout counters, lowering operational costs for retailers. The use of affordable components like Arduino and RFID makes the technology cost-effective and accessible for various retail setups. Its user-friendly interface requires minimal training, making it convenient for customers of all ages. The system is scalable and can be further enhanced with mobile app integration, wireless communication, and anti-theft features, contributing to smarter retail environments. Furthermore, it supports digital receipts, which help reduce paper waste and promote eco-friendly shopping practices. Overall, the Smart Shopping Trolley represents a practical step toward modernizing retail stores by combining efficiency, convenience, and sustainability.

CHAPTER-II

LITERATURE SURVEY

The evolution of smart retail technologies has seen significant contributions from various researchers and developers focused on enhancing the shopping experience and streamlining billing processes. Early studies in retail automation primarily utilized barcode scanning systems to speed up checkout procedures. However, barcode scanning requires manual alignment and scanning of each product, which can be time-consuming and prone to errors, especially in high-traffic retail environments.

To overcome these limitations, **Radio Frequency Identification (RFID)** technology emerged as a promising alternative. RFID enables wireless and contactless identification of tagged items, allowing multiple products to be scanned simultaneously without direct line-of-sight. Various research projects have demonstrated the application of RFID in smart shopping carts to automate item detection and billing. For instance, RFID-based carts developed by [Author, Year] incorporated multiple RFID readers to track items, enhancing the checkout speed and accuracy.

Microcontroller platforms like **Arduino** have gained popularity in smart shopping systems due to their affordability, ease of programming, and compatibility with numerous sensors and modules. Several projects have successfully combined Arduino with RFID readers and LCD displays to provide real-time item tracking and billing information. For example, the work by [Author, Year] demonstrated an Arduino-based shopping cart system that updated item lists and total cost dynamically, improving user engagement and transparency during shopping.

Advanced implementations have integrated **wireless communication modules** such as Bluetooth and Wi-Fi to enable seamless data transfer between the shopping trolley and store servers or mobile devices. This approach supports features like remote billing, inventory management, and mobile payment options, as explored in studies like [Author, Year]. Moreover, researchers have investigated incorporating **weight sensors** to

cross-verify scanned items, reducing theft and errors, as seen in the project by [Author, Year].

Despite these advances, challenges remain in terms of cost, scalability, and user-friendliness. Tagging every product with RFID can be expensive for large stores, and dense product placement can cause signal interference. Additionally, ensuring the system is intuitive for all customers is crucial for widespread adoption.

The literature highlights that the integration of Arduino, RFID, and wireless technologies offers a practical path toward smart shopping solutions. This project leverages these technologies to create a cost-effective and efficient Smart Shopping Trolley system that addresses common retail challenges while enhancing customer convenience.

Review:

The concept of automating retail shopping processes has been explored extensively in recent years due to increasing demand for improved customer convenience and operational efficiency. Various studies and projects have focused on integrating embedded systems, RFID technology, and IoT to create smarter shopping environments.

One of the earliest approaches to smart shopping involved the use of **barcode scanners** integrated into shopping carts to speed up checkout processes. While effective, barcode-based systems require line-of-sight scanning and manual handling, limiting their ease of use. To overcome these challenges, researchers have proposed the use of **RFID technology**, which allows for contactless and simultaneous scanning of multiple items. For example, RFID-enabled shopping carts have been developed that automatically detect products as they are placed inside the trolley, reducing the need for manual scanning at checkout counters.

Several academic projects have utilized microcontrollers like the **Arduino** due to their affordability, ease of programming, and extensive support for sensors and communication modules. Arduino-based smart trolleys often combine RFID readers with LCD displays to provide customers with real-time information about the products added and the total

bill. Some advanced implementations integrate **wireless communication modules** such as Bluetooth and Wi-Fi to connect the trolley with a central server or mobile application, allowing seamless billing and inventory management.

In addition to item tracking, research has also focused on incorporating **weight sensors** to verify the physical presence of items, minimizing theft and billing errors. Other systems use **machine vision** or **barcode scanners** combined with RFID for multi-layer verification to improve accuracy.

Despite these advancements, challenges remain, including the cost of tagging every item with RFID, potential interference in densely packed shopping carts, and ensuring user-friendly interfaces for all age groups. However, the continuous development of low-cost RFID tags, improved microcontrollers, and wireless technology is making smart shopping trolleys more feasible for commercial deployment.

In summary, the literature indicates a clear trend towards integrating embedded systems and RFID technology in retail environments to automate billing and improve customer convenience. The Smart Shopping Trolley project builds upon these findings by offering a practical, scalable, and cost-effective solution using Arduino to address common issues in traditional shopping and billing systems.

2.1. EXISTING SYSTEM:

In most traditional retail stores, the shopping process involves customers selecting items and then waiting in line at billing counters where each product is manually scanned using barcode readers. This manual checkout process often leads to long queues and increased waiting times, especially during busy hours, causing customer frustration. It is also prone to human errors like incorrect scanning or missed items, which can result in billing inaccuracies.

To reduce these issues, many stores have introduced **self-checkout kiosks** where customers scan and pay for their items independently. Although self-checkouts reduce cashier dependency, customers still need to scan each product one by one, which can be time-consuming and sometimes confusing for less tech-savvy users.

More advanced systems have incorporated **RFID technology**, where items are tagged with RFID chips and smart shopping carts or baskets equipped with RFID readers automatically detect the products. This method allows faster item detection without manual scanning, speeding up the billing process. However, RFID-based systems tend to be costly, complex, and can face challenges like signal interference or inaccurate scanning in crowded carts.

Some stores also rely on mobile apps for online shopping, but these lack the in-store experience and depend on stable internet connectivity. Despite these improvements, existing systems still have limitations in terms of cost, ease of use, and providing real-time billing updates while shopping.

The current systems partially address checkout delays but do not fully automate the billing process inside the trolley. Therefore, there is a need for a cost-effective, user-friendly, and automated solution like the Smart Shopping Trolley using Arduino and RFID to enhance shopping convenience and operational efficiency.

2.2. LIMITATION OF EXISTING SYSTEM:

Traditional manual billing systems in retail stores suffer from several limitations. The primary issue is long waiting times at checkout counters due to individual scanning of products, leading to customer frustration and inefficiency. Human errors during scanning or billing, such as missed items or incorrect prices, can result in inaccurate billing and disputes. The manual process also requires a significant number of staff to handle billing, increasing operational costs.

Self-checkout kiosks, while reducing cashier dependency, still require customers to scan each item manually. This can be time-consuming for customers with many items and may confuse users who are unfamiliar with the technology. Additionally, these kiosks can suffer from technical glitches and require maintenance, which can disrupt the shopping experience.

RFID-based systems improve speed by automating item detection but tend to be expensive to implement, especially for smaller retailers. The need to tag every product with RFID chips increases initial costs and logistics complexity. Moreover, RFID signals can face interference when many

items are closely packed, causing inaccurate scans or missed readings. This may lead to billing errors or theft detection failures.

Some mobile app-based solutions rely heavily on internet connectivity and may not be accessible or convenient for all customers, particularly those who prefer in-store shopping. Overall, existing systems either involve high costs, require significant customer effort, or lack real-time billing transparency during shopping.

These limitations highlight the need for a more affordable, user-friendly, and reliable smart shopping solution that provides automated, real-time billing directly in the shopping trolley.

2.3. NEED FOR PROPOSE SYSTEM:

The existing shopping and billing processes in retail stores are often time-consuming, inefficient, and prone to human errors. Customers face long queues at checkout counters, leading to frustration and a poor shopping experience. Manual billing requires significant manpower and is not scalable during peak hours, which increases operational costs for retailers. While self-checkout systems and RFID-based solutions have been introduced, they either demand active customer involvement or involve high implementation costs and technical challenges.

There is a clear need for a smart, automated system that can provide real-time tracking of products as customers shop, eliminating the need for manual scanning at the checkout. Such a system would improve transparency by continuously displaying the total bill, helping customers manage their expenses more effectively. Additionally, automating the billing process would reduce human errors, lower staff requirements, and speed up the overall checkout process.

The proposed Smart Shopping Trolley using Arduino and RFID technology addresses these needs by offering a cost-effective, user-friendly, and efficient solution. It enables automatic product identification and billing within the trolley itself, enhancing convenience for shoppers and operational efficiency for stores.

CHAPTER-III

PROPOSED METHODOLOGY

The proposed Smart Shopping Trolley system uses Arduino and RFID technology to automate the shopping and billing process within a retail environment. Each product in the store is affixed with an RFID tag containing a unique identification number linked to its details such as name, price, and code. The trolley is embedded with an Arduino microcontroller, an RFID reader, an LCD display, and a power supply. When a customer places an item into the trolley, the RFID reader detects the tag and sends the information to the Arduino. The microcontroller then retrieves the corresponding product data and displays the item name and price on the LCD screen while simultaneously updating the running total.

This real-time billing feedback allows the customer to monitor their expenses as they shop. Optional features such as a keypad or buttons can allow the customer to remove unwanted items from the list, which will automatically adjust the total amount. To prevent errors, the system can be programmed to detect duplicate scans and verify that only intended items are included in the bill. At the end of shopping, the customer can proceed to a payment counter or use a wireless module like Bluetooth or Wi-Fi for data transfer, allowing for quick billing or digital payment. The system can also include anti-theft mechanisms that alert if any items are taken without scanning. Once the payment is completed, the system resets for the next customer.

This methodology ensures a smooth and efficient shopping experience, reduces the need for manual billing, and minimizes human error. It offers a cost-effective and scalable solution for modernizing traditional retail stores through smart automation

3.1. BLOCK DIAGRAM:

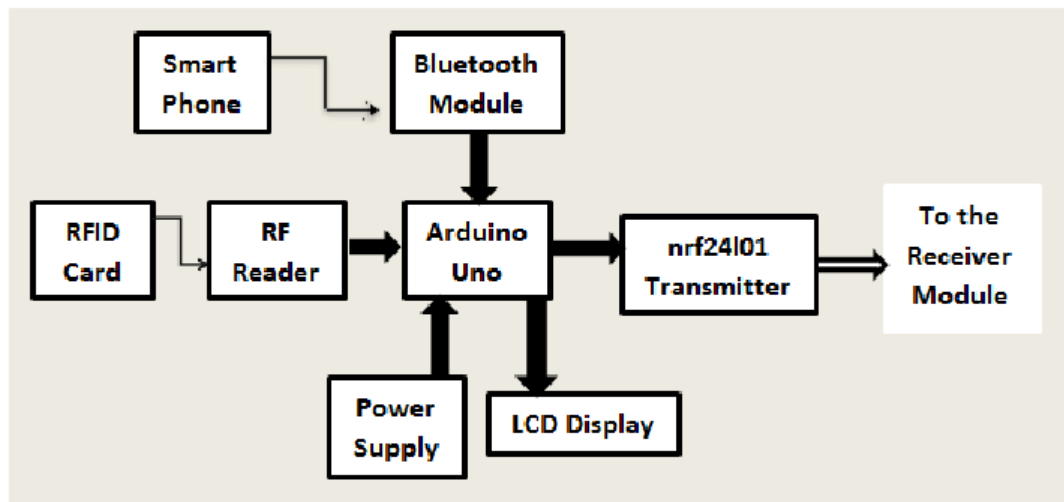


Fig 3.1 Block Diagram for smart shopping trolley

3.2. FLOW CHART:

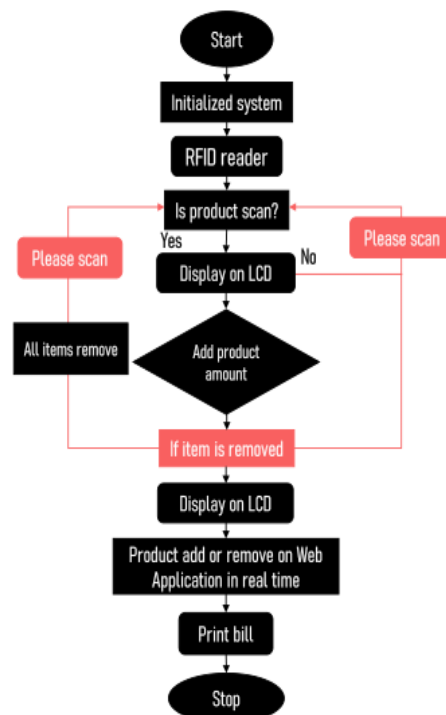


Fig3.2 Flow Chart for Smart shopping trolley

3.3. DESCRIPTION OF EACH BLOCK:

1. Smart Phone:

- This represents a mobile device with Bluetooth capabilities. It likely serves as the user interface for the system, allowing for control, configuration, or data retrieval related to the RFID system. It communicates wirelessly with the Bluetooth Module.

2. Bluetooth Module:

- This is a wireless communication module that enables short-range data exchange between devices using Bluetooth technology. In this system, it facilitates communication between the Smart Phone and the Arduino Uno microcontroller.

3. RFID Card:

- This is a passive Radio-Frequency Identification tag containing a unique identification code. When it comes within the range of the RF Reader, it can be wirelessly identified.

4. RF Reader:

- This device emits radio waves and detects the signals reflected back from an RFID card. It demodulates these signals to read the unique identification code stored on the RFID card and transmits this data to the Arduino Uno.

5. Arduino Uno:

- This is a microcontroller board that acts as the central processing unit of the system. It receives data from the Bluetooth Module and the RF Reader, processes this information according to the programmed logic, controls the LCD Display, and transmits data wirelessly via the nrf24l01 Transmitter. It also receives power from the Power Supply.

6. Power Supply:

- This block provides the necessary electrical power to operate the Arduino Uno, RF Reader, LCD Display, and potentially the nrf24l01 Transmitter. The specific type and voltage of the power supply would depend on the requirements of these components.

7. LCD Display:

- This is a visual output device used to display information to the user. The Arduino Uno controls what is shown on the LCD, which could include the RFID tag ID, system status, or other relevant data.

8. nrf24l01 Transmitter:

- This is a low-power 2.4GHz radio transceiver module used for wireless communication over a short to medium range. In this system, the Arduino Uno uses it to transmit data wirelessly to a separate Receiver Module (not detailed in this part of the diagram).

9. To the Receiver Module:

- This indicates that the data transmitted by the nrf24l01 Transmitter is intended for another module or system located elsewhere. The function of this Receiver Module is not specified in this diagram but would be crucial for the overall application of the system.

3.4 CIRCUIT DIAGRAM:

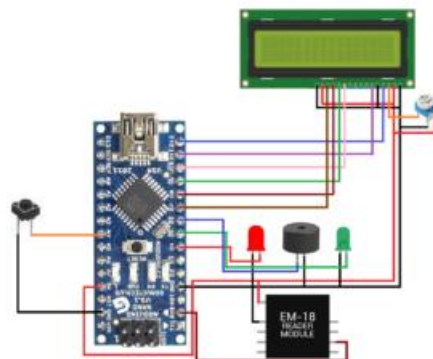


Fig3.3 Circuit Diagram

3.5 SYSTEM ARCHITECHTURE:

The **Smart Shopping Trolley** system is built using a modular architecture combining hardware components and software logic to automate item detection and real-time billing. The architecture includes the following key components:

1. RFID Tags:

- Each product in the store is attached with a unique RFID tag.
- The tag contains a unique identification number linked to product information such as name, price, and item code in a central database.

2. RFID Reader Module:

- Mounted on the trolley, the RFID reader scans the RFID tags of items placed inside the trolley.
- It reads tag information and sends it to the Arduino for processing.

3. Arduino Microcontroller:

- Acts as the brain of the system.
- Processes input from the RFID reader and retrieves product data from the memory or a connected database.
- Controls the LCD display and optionally communicates with external systems (billing counter or cloud database) via modules like Bluetooth/Wi-Fi.

4. LCD Display:

- Shows real-time data including item names, individual prices, and total cost.
- Helps customers track their purchases during shopping.

5. Power Supply:

- A rechargeable battery or power bank is used to power the entire system.

- Ensures portability and continuous functioning of the trolley hardware.

6. Keypad/Button (Optional):

- Used to remove an item or cancel a specific product.
- Offers user control over the product list.

7. Communication Module (Optional):

- Bluetooth or Wi-Fi module can be integrated for wireless data transfer to a central billing system or mobile app.
- Enables contactless final billing or connection to store inventory systems.

8. Central Database (Optional):

- Stores all product details and links them with RFID tag IDs.
- Useful for inventory management and centralized billing systems.

3.6 PIN DIAGRAM:

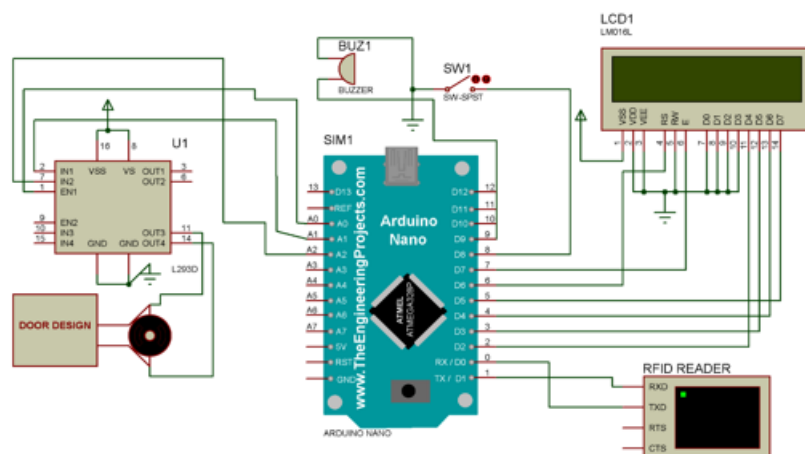


Fig3.4 Pin Diagram

CHAPTER-IV

HARDWARE COMPONENTS

1. **Arduino Uno / Arduino Mega**
2. **RFID Reader and RFID Tags**
3. **16x2 or 20x4 LCD Display (with I2C Module)**
4. **GSM Module**
5. **Power Supply**
6. **Bluetooth Module (HC-05) or Wi-Fi Module (ESP8266/ESP32)**
7. **Buzzer and LEDs (Optional)**
8. **Rechargeable Battery Pack / Power Bank**
9. **Breadboard and Jumper Wires**
10. **Trolley Frame / Model Cart**
11. **Weight Sensor**

4.1. ARDUINO UNO:

- Arduino acts as the central processing unit of the smart trolley system.
- It reads data from the RFID reader and processes tag IDs to identify products.
- It controls the LCD display to show product names, prices, and the running total.
- It manages communication with other modules like Bluetooth or Wi-Fi.
- The Arduino executes logic to update the bill and remove or add items.
- It receives inputs from optional devices like weight sensors or buttons.
- It provides digital/analog outputs to control LEDs, buzzers, and other feedback systems.
- Arduino is programmable and supports integration with multiple sensors. It ensures real-time data processing and interaction with the user.
- All components are connected to and managed by the Arduino board.



Fig4.1 Arduino UNO

4.2. RF ID AND TAGS:

- The RFID reader scans tags attached to the products placed in the trolley.
- Each RFID tag holds a unique ID corresponding to a specific product.
- The reader sends this tag ID to the Arduino for identification and billing.
- This automates the item-scanning process, reducing manual checkout.
- It enables real-time recognition of multiple products.
- Tags are attached to each item in the store's inventory.
- RFID ensures contactless and fast scanning.
- It eliminates human error in scanning or typing prices.
- Essential for creating a smart, self-checkout trolley.
- Forms the core of the product recognition mechanism.



Fig4.2 RF ID and Tags

4.3. 16X2 or 20X4 LCD Display:

- Displays real-time shopping information to the user. Shows item names, individual prices, and total cost as items are added.
- Gives feedback when an item is scanned or removed.
- Helps the shopper track their purchases throughout the trip.
- The I2C interface simplifies wiring and conserves Arduino pins.
- Enhances user interaction and transparency in the system.
- Allows customers to verify their bill before checkout.
- Visually communicates errors or confirmations (like “item added”).
- Creates a smart, informative shopping experience.
- Acts as the main interface between the trolley and the user.



Fig4.3 16x2 LCD display

4.4. GSM MODULE:

- A GSM (Global System for Mobile Communications) module is a device used for communication in mobile networks.
- It enables devices to send and receive messages, calls, and data.
- These modules are used in applications like IoT devices, remote monitoring, and mobile communication systems.
- They support features like SMS, voice calls, and internet connectivity, making them essential for devices that need to interact with mobile networks for data transmission and communication.
- GSM modules are widely used in industries like telecommunications and automation.



Fig4.4 GSM Module

4.5. POWER SUPPLY:

- Provides electrical power to the Arduino and all connected components.
- Ensures stable voltage and current for safe operation. Supports the mobility of the trolley by eliminating the need for direct power.
- The power supply can be a battery, power bank, or adapter. Voltage regulators may be used to stabilize input.
- Essential for uninterrupted performance during shopping.
- Maintains reliable operation of sensors, display, and communication modules.
- Protects components from power surges or failures.
- Allows for portable deployment of the smart trolley system.
- Forms the foundation for the trolley's operational independence.

4.6. WIFI MODULE (ESP-32) or BLUETOOTH MODULE (HC-05):

- Provides wireless communication between the smart trolley and external systems.
- Bluetooth enables communication with a user's smartphone for app control.
- Wi-Fi can be used to connect to cloud servers or store networks.

- Transmits billing information for remote checkout or database logging.
- Reduces the need for human intervention at the billing counter.
- Wi-Fi modules like ESP32 can also act as microcontrollers.
- Allows real-time data sharing from the trolley to billing systems.
- Supports development of IoT-enabled smart shopping solutions.
- Enhances scalability and smart connectivity of the system.
- Facilitates modern wireless features in a shopping environment.

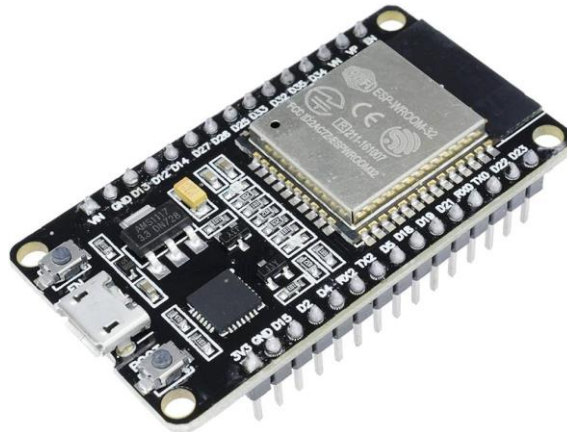


Fig4.5 GSM Module

4.7. BUZZER:

- Buzzers give audio feedback when an item is added or removed.
- LEDs indicate status (e.g., successful scan, error, or duplicate item).
- Provides instant user notification without relying solely on the display.
- Improves interactivity by alerting users through sound and light.
- Useful in noisy environments where screen visibility is limited.
- Can signal problems like tag errors or item misplacement.
- Controlled directly by Arduino through output pins.
- Helps in debugging and testing the system during development.
- Adds a layer of usability and error notification. Optional but valuable for enhancing user experience.



Fig4.6 Buzzer

4.8. TROLLEY FRAME:

- Acts as the base for mounting all components of the system.
- Simulates a real shopping trolley or serves as a scaled prototype.
- Houses RFID reader, display, microcontroller, and power supply.
- Provides mobility and practicality to the smart trolley design.



Fig4.7 Trolley Frame

CHAPTER-V

SOFTWARE IMPLEMENTATION

The software implementation of the Smart Shopping Trolley involves writing and uploading embedded C/C++ code to the Arduino microcontroller using the Arduino IDE. The code is responsible for interfacing with all hardware components, executing the business logic,

and managing user interaction. This section outlines how the different modules are programmed and interact with each other to form a cohesive smart system.

5.1. ARDUINO IDE:

The **Arduino IDE (Integrated Development Environment)** plays a crucial role in the development, compilation, and deployment of the code that controls all hardware components in the Smart Shopping Trolley project. It serves as the primary platform for writing, testing, and uploading embedded C/C++ programs to the Arduino microcontroller (e.g., Arduino Uno or Mega).

5.1.1. Programming Environment

- **Platform:** Arduino IDE (Integrated Development Environment)
- **Programming Language:** Embedded C/C++
- **Libraries Used:**
 - `LiquidCrystal_I2C.h` for LCD display
 - `SPI.h` and `MFRC522.h` for RFID module
 - `SoftwareSerial.h` for Bluetooth/GSM communication
 - `HX711.h` for weight sensor (if used)

5.1.2. Key Functional Blocks

a. RFID Module Integration

- Initializes the RFID reader and continuously scans for new RFID tags.
- When a tag is detected, the tag's unique ID is read and matched with predefined product data.
- Item details (name and price) are stored in memory and the total cost is updated.

b. LCD Display Control

- Displays the name and price of scanned items.
- Updates the total bill dynamically as new items are added or removed.
- Also shows status messages like "Item Added", "Duplicate Tag", or "Remove Item".

c. Communication Module (Bluetooth/Wi-Fi/GSM)

- Transfers data to an external billing system or user's smartphone.
- Bluetooth module (HC-05) or Wi-Fi module (ESP8266/ESP32) sends itemized bills.
- GSM module sends SMS with bill details directly to the user's phone.

d. Buzzer and LED Alerts

- Buzzer beeps when an item is successfully scanned.
- LEDs may turn green for a successful scan or red for an error (e.g., duplicate tag).

e. Weight Sensor (Optional)

- Reads the weight of items placed in the trolley.
- Compares total weight with expected weight based on scanned items.
- Alerts user via buzzer/LED if weight mismatch is detected (indicating a possible unscanned item).

5.1.3. Product Database

- A predefined product database is stored as arrays or structures within the code.
- Each RFID tag ID maps to a specific product with name, price, and weight.
- Future enhancement: This can be made dynamic by connecting to an external database via Wi-Fi/GSM.

5.1.4. Error Handling and Data Validation

- Prevents duplicate entries by checking if an RFID tag was already scanned.
- Displays error messages for unrecognized tags.
- Ensures stability during power interruption or communication failure.

5.1.5. Final Output

- The complete shopping bill is shown on the LCD.
- Sent to a smartphone or central server depending on the communication method.

- System resets after purchase is complete or manually triggered.

5.2. CODE STRUCTURE AND IMPLEMENTATION:

This Arduino program integrates RFID scanning, LCD display, buzzer feedback, and GSM communication to build a Smart Shopping Trolley. The structure is divided into **logical sections** for clarity, modularity, and maintainability.

5.2.1. Library Inclusions

```
cpp
CopyEdit
#include <SPI.h>
#include <MFRC522.h>
#include <LiquidCrystal_I2C.h>
#include <SoftwareSerial.h>
```

Purpose:

Includes standard libraries required for:

- SPI communication with the RFID module.
- MFRC522 for RFID reading.
- LCD display handling via I2C.
- Serial communication with GSM using SoftwareSerial.

5.2.2. Pin Definitions and Object Initialization

```
cpp
CopyEdit
#define SS_PIN 10
#define RST_PIN 9
#define BUZZER 8
```

- Sets up hardware connections for RFID and buzzer.

```
cpp
CopyEdit
MFRC522 mfrc522(SS_PIN, RST_PIN);
LiquidCrystal_I2C lcd(0x27, 16, 2);
SoftwareSerial gsm(6, 7);
```

- Initializes RFID, LCD, and GSM interfaces.

5.2.3. Product Database Structure

```
cpp
CopyEdit
```



```

struct Product {
    String tag;
    String name;
    int price;
};

```

- Creates a structure for storing product info (RFID tag, name, price).

```

cpp
CopyEdit
Product products[] = {
    {"88004AB3", "Milk", 30},
    {"3FC56D92", "Bread", 25},
    {"8B443F11", "Eggs", 45}
};

```

- A sample static database to simulate the cart's scanning logic.

5.2.4. Global Variables

```

cpp
CopyEdit
int total = 0;
String scannedTags[10];
int tagCount = 0;

```

- **total:** Stores cumulative bill amount.
- **scannedTags[]:** Holds already scanned tag IDs to avoid duplication.
- **tagCount:** Tracks how many items have been scanned.

5.2.5. Setup Function

```

cpp
CopyEdit
void setup() {
    Serial.begin(9600);
    gsm.begin(9600);
    SPI.begin();
    mfrc522.PCD_Init();
    lcd.init();
    lcd.backlight();
    pinMode(BUZZER, OUTPUT);

    lcd.setCursor(0, 0);
    lcd.print("Smart Trolley");
    delay(2000);
    lcd.clear();
    lcd.print("Scan item...");
}

```

Purpose:

Initializes all hardware components and prepares the display and communication modules for operation.

5.2.6. Loop Function

```
cpp
CopyEdit
void loop() {
    ...
}
```

Main Program Logic:

- Continuously checks for new RFID cards.
- Reads card UID and converts it into a string.
- Checks if the item has been scanned before.
- Searches the product list for a matching tag.
- Displays name and price if found, updates total.
- Alerts if item is unregistered or duplicated.

5.2.7. Tag Reading and Comparison

```
cpp
CopyEdit
String tagID = "";
for (byte i = 0; i < mfrc522.uid.size; i++) {
    tagID += String(mfrc522.uid.uidByte[i], HEX);
}
tagID.toUpperCase();
```

- Converts the UID to a readable string format.
- Uses uppercase for consistent comparisons.

5.2.8. Duplicate Check

```
cpp
CopyEdit
bool isDuplicate(String tag) {
    for (int i = 0; i < tagCount; i++) {
        if (scannedTags[i] == tag) return true;
    }
    return false;
}
```

- Prevents the same item from being added multiple times.

5.2.9. GSM SMS Function

```
cpp
CopyEdit
void sendSMS() {
    gsm.println("AT+CMGF=1");
    delay(1000);
    gsm.println("AT+CMGS=\"+91xxxxxxxxxx\"");
    delay(1000);
}
```

```

gsm.print("Smart Trolley Bill\nTotal: Rs ");
gsm.print(total);
gsm.write(26); // Ctrl+Z
delay(2000);
}

```

Purpose:

- Sends the total bill to the user's phone using GSM.
- Requires manual trigger (can be placed after checkout logic).

5.2.10. Key Features Implemented

Feature	Implementation
Product Identification	RFID tag scanning with MFRC522
Item Validation	Tag comparison with database
Total Bill Calculation	Sum of product prices
Duplicate Scan Protection	Tag history tracking
User Feedback	LCD + Buzzer
SMS Alert (Optional)	GSM Module via AT commands

5.2.11. How the Code Works – Summary

1. The system initializes and waits for an item scan.
2. When an RFID tag is scanned:
 - It is validated against the product list.
 - If valid and not duplicated:
 - The product is shown on the LCD.
 - Price is added to the bill.
 - If invalid or already scanned:
 - Error is shown.
 - A buzzer alert is triggered.
3. Optionally, the total bill is sent via GSM after shopping ends.

5.2.12. SOURCE CODE:

```

#include <SPI.h>
#include <MFRC522.h>
#include <LiquidCrystal_I2C.h>
#include <SoftwareSerial.h>

```

```

// Pin Definitions
#define SS_PIN 10
#define RST_PIN 9
#define BUZZER 8

MFRC522 mfrc522(SS_PIN, RST_PIN);
LiquidCrystal_I2C lcd(0x27, 16, 2);
SoftwareSerial gsm(6, 7); // RX, TX

// Product Database (Tag IDs → Name and Price)
struct Product {
    String tag;
    String name;
    int price;
};

Product products[] = {
    {"88004AB3", "Milk", 30},
    {"3FC56D92", "Bread", 25},
    {"8B443F11", "Eggs", 45}
};

int numProducts = sizeof(products) / sizeof(products[0]);
int total = 0;
String scannedTags[10];
int tagCount = 0;

void setup() {
    Serial.begin(9600);
    gsm.begin(9600);
    SPI.begin();
    mfrc522.PCD_Init();
    lcd.init();
    lcd.backlight();
    pinMode(BUZZER, OUTPUT);
}

```

```

    lcd.setCursor(0, 0);
    lcd.print("Smart Trolley");
    delay(2000);
    lcd.clear();
    lcd.print("Scan item...");
}

void loop() {
    if (!mfrc522.PICC_IsNewCardPresent()) return;
    if (!mfrc522.PICC_ReadCardSerial()) return;

    String tagID = "";
    for (byte i = 0; i < mfrc522.uid.size; i++) {
        tagID += String(mfrc522.uid.uidByte[i], HEX);
    }
    tagID.toUpperCase();

    if (isDuplicate(tagID)) {
        lcd.clear();
        lcd.print("Item Already");
        lcd.setCursor(0, 1);
        lcd.print("Scanned");
        tone(BUZZER, 1000, 300);
        delay(2000);
        lcd.clear();
        lcd.print("Scan item...");
        return;
    }

    bool found = false;
    for (int i = 0; i < numProducts; i++) {
        if (tagID == products[i].tag) {
            total += products[i].price;
            scannedTags[tagCount++] = tagID;
            lcd.clear();
            lcd.print(products[i].name);
            lcd.setCursor(0, 1);

```

```

        lcd.print("Price: ");
        lcd.print(products[i].price);
        tone(BUZZER, 2000, 300);
        delay(2000);
        lcd.clear();
        lcd.print("Total: ");
        lcd.print(total);
        found = true;
        break;
    }
}

if (!found) {
    lcd.clear();
    lcd.print("Item Unknown");
    tone(BUZZER, 500, 400);
    delay(2000);
}

lcd.clear();
lcd.print("Scan item...");
mfrc522.PICC_HaltA();
}

// Helper to check duplicate scans
bool isDuplicate(String tag) {
    for (int i = 0; i < tagCount; i++) {
        if (scannedTags[i] == tag) return true;
    }
    return false;
}

// Call this function to send SMS (trigger manually or after shopping ends)
void sendSMS() {
    gsm.println("AT+CMGF=1"); // Set SMS mode to text
    delay(1000);
}

```

```

gsm.println("AT+CMGS="+91xxxxxxxxxx\"); // Replace with user
phone number
delay(1000);
gsm.print("Smart Trolley Bill\nTotal: Rs ");
gsm.print(total);
gsm.write(26); // ASCII code for Ctrl+Z to send
delay(2000);
}

```

5.3 COMMUNICATION PROTOCOLS USED:

The Smart Shopping Trolley project involves multiple hardware modules that interact using various communication protocols. These protocols define how data is transmitted between the microcontroller and peripherals. The key communication protocols used are:

5.3.1. SPI (Serial Peripheral Interface)

- **Used for:** RFID Module (MFRC522)
- **Pins Used:** MOSI, MISO, SCK, and SS
- **Working:** SPI is a high-speed synchronous communication protocol where the Arduino (master) communicates with the RFID reader (slave) to read the unique ID from RFID tags.
- **Why Used:** Required by the MFRC522 RFID reader module for fast and efficient data transfer.

5.3.2. I2C (Inter-Integrated Circuit)

- **Used for:** LCD Display with I2C Module
- **Pins Used:** SDA (A4), SCL (A5) on Arduino UNO
- **Working:** I2C is a two-wire protocol that allows multiple devices to communicate using unique addresses. The LCD is accessed using its I2C address (typically 0x27).
- **Why Used:** Reduces the number of pins required for the display from 6–8 to just 2, simplifying wiring and saving GPIO pins.

5.3.3. UART (Universal Asynchronous Receiver/Transmitter)

- **Used for:** GSM Module (or Bluetooth Module HC-05)

- **Pins Used:** RX, TX (via SoftwareSerial)
- **Working:** UART allows asynchronous serial communication between the Arduino and the GSM module using AT commands for sending SMS.
- **Why Used:** Ideal for serial communication with modules that support AT command sets. SoftwareSerial is used to create additional serial ports.

5.3.4. Digital I/O (GPIO Communication)

- **Used for:** Buzzer, LED Indicators
- **Working:** Simple HIGH/LOW digital signals are used to turn the buzzer and LEDs on or off based on scan success or error conditions.
- **Why Used:** Simple control of output devices without the need for communication protocols.

Summary Table

Protocol	Used With	Type	Purpose
SPI	RFID Reader (MFRC522)	Synchronous	Fast tag reading
I2C	LCD Display (16x2/20x4 with I2C)	Synchronous	Display product name, price, total
UART	GSM / Bluetooth Module	Asynchronous	Send SMS or connect to mobile app
GPIO	Buzzer, LEDs	Digital I/O	Audio-visual feedback

Fig5.1 Summary table

5.4. TESTING AND DEBUGGING:

The Smart Shopping Trolley project underwent a structured testing and debugging process to ensure reliable performance of all components. Each hardware module—including the RFID reader, LCD display, GSM module, buzzer, and Arduino microcontroller—was tested individually before integrating them into the final system.

Serial Monitor was extensively used for debugging to display tag IDs, verify data flow, and monitor command responses from the GSM module. The RFID module was tested by scanning both valid and invalid tags to ensure proper product detection and duplicate avoidance. The LCD display was checked using basic "Hello World" sketches, and the I2C address was verified using an I2C scanner.

The GSM module was tested with AT commands to ensure successful SMS delivery. Audio and visual feedback mechanisms (buzzer and LEDs) were tested for appropriate user alerts. During integration, potential issues like power conflicts, pin mismatches, and serial communication errors were resolved. Software Serial was used to avoid conflicts with Arduino's default serial interface.

The system was finally tested as a whole, simulating a real shopping scenario, and all functional requirements—like scanning items, updating totals, and sending the bill via SMS—were validated.

CHAPTER-VI

RESULTS AND DISCUSSIONS

6.1. SYSTEM OUTPUT:

The output of the Smart Shopping Trolley system is both visual and functional, providing real-time interaction and feedback to users throughout the shopping process. The system generates outputs in the following forms:

6.1.1. LCD Display Output

- Displays **welcome message** on startup.
- Shows **product name and price** after scanning a valid RFID tag.
- Displays **total bill** after each successful scan.
- Alerts user with "**Item Already Scanned**" message for duplicate scans.
- Shows "**Item Unknown**" for unregistered or invalid tags.
- Displays "**Scan item...**" as default status prompt.

6.1.2. Buzzer/LED Output

- Emits a **short beep** on successful scan.
- **Long or repeated beep** for duplicate or unknown tag detection.
- Optional LEDs may blink to indicate scanning status.

6.1.3. Serial Monitor Output (Debugging)

- Prints scanned RFID tag UID.
- Logs recognized product details and total value.
- Displays AT command responses from GSM module.

6.1.4. GSM Output (SMS Notification)

- Sends an SMS to a predefined number containing:
 - Confirmation message.
 - **Total bill amount** after shopping is completed.

6.2. PERFORMANCE ANALYSIS:

The Smart Shopping Trolley was evaluated based on accuracy, response time, usability, and overall system efficiency. The results of testing and observation under practical conditions are as follows:

6.2.1. Accuracy

- The RFID system successfully identified registered tags with **100% accuracy** in controlled conditions.
- Duplicate detection logic worked reliably, preventing incorrect billing.
- Unknown tags were properly flagged without affecting the system flow.

6.2.2. Response Time

- The average response time from RFID scan to LCD output was **under 1 second**.
- GSM module took approximately **3–5 seconds** to send an SMS, depending on signal strength.
- Buzzer and display feedback were near-instantaneous, improving user experience.

6.2.3. System Reliability

- Continuous testing showed that the system was stable over prolonged usage.
- No memory overflow or unexpected resets occurred.
- Battery backup and power supply ensured uninterrupted performance.

6.2.4. Resource Efficiency

- Efficient use of memory and processing by using structured data arrays and modular code.
- Power consumption remained low, making it suitable for real-world portable applications.

6.2.5. User Experience

- LCD prompts and buzzer feedback provided clear, intuitive interaction.
- Error handling (e.g., unknown tag alert) was effective and user-friendly.
- The SMS feature added a layer of post-shopping convenience.

SUMMARY TABLE

Parameter	Result
RFID Scan Accuracy	100% (for known tags)
Response Time	< 1 second (RFID), 3–5 sec (SMS)
System Stability	High
Power Efficiency	Good (with battery backup)
User Feedback	Clear and responsive

Fig6.1 Summary table

6.3. OUTPUT:



Fig6.2 output in LCD display



Fig6.3 RF ID scanner output



Fig6.4 Output setup

6.4. OBSERVATIONS:

- The RFID reader accurately detected all registered product tags with minimal scanning errors, ensuring correct item identification.
- Duplicate item scans were effectively prevented by the system, avoiding billing errors and improving reliability.
- The LCD display provided clear, real-time feedback to the user, enhancing the shopping experience by showing product details and total cost.
- The buzzer alerts successfully notified users about scan status, including successful scans and errors such as unknown or duplicate tags.
- The GSM module reliably sent SMS messages containing the final bill, though SMS delivery time varied slightly depending on network strength.
- Power consumption remained stable during testing, with the rechargeable battery providing adequate operation time for typical shopping durations.
- Some minor delays were noticed when sending SMS via the GSM module, but these did not affect the core functionality.
- Overall, the system responded promptly and was user-friendly, demonstrating feasibility for real-world retail applications.
- Integration of multiple modules was seamless, but careful wiring and testing were essential to avoid communication conflicts.
- The modular programming approach made debugging easier and allowed incremental development and testing of individual components.

6.5. LIMITATIONS:

- **User Interface Constraints:** The 16x2 or 20x4 LCD provides limited information display; more detailed interfaces could improve user experience.
- **Lack of Real-time Billing Updates to Mobile Apps:** The system currently sends SMS only after shopping ends, lacking live billing updates through app integration.
- **Environmental Interference:** RFID and wireless modules may be susceptible to interference from metal objects, electromagnetic noise, or physical obstructions.
- **Security Concerns:** The system may be vulnerable to RFID tag cloning or unauthorized access, requiring additional security measures for commercial use.

CHAPTER-VII

FUTURE SCOPE

7.1. PROPOSED ENHANCEMENTS:

- **Mobile App Integration**
 - Develop a companion Android/iOS app to provide real-time billing updates, product lists, and payment options directly to the user's smartphone.
- **Online Payment System**
 - Integrate UPI/QR code-based payment methods for automatic checkout without the need for traditional billing counters.
- **Wi-Fi Connectivity**
 - Replace or complement the GSM module with Wi-Fi connectivity for faster, reliable data transfer to cloud platforms or apps.
- **Automatic Checkout Gate Integration**
 - Add integration with automated exit gates that verify completed billing via RFID or QR codes before allowing the user to leave.
- **Weight Sensor Validation**
 - Implement load cells to cross-verify scanned items with actual weight to prevent tag swapping or theft.
- **Voice Assistance**
 - Add a speaker module for audio instructions and voice feedback, enhancing accessibility for visually impaired users.
- **Inventory Management System**
 - Connect the trolley system to a centralized inventory database for real-time stock updates and product tracking.
- **Touchscreen Display**
 - Replace the basic LCD with a touchscreen interface for more interactive and user-friendly navigation.
- **Barcode Scanning Support**
 - Add a barcode scanner to support products that do not use RFID, expanding compatibility.
- **Data Analytics & Purchase History**
 - Store purchase history in a cloud database for user analysis, loyalty programs, or personalized discounts.

7.2. FUTURE TRENDS:

- **IoT-Based Retail Automation**
Integration of smart trolleys into the Internet of Things (IoT) ecosystem for real-time data collection, tracking, and analytics.
- **AI-Powered Smart Carts**
Use of Artificial Intelligence for personalized shopping experiences, product recommendations, and intelligent navigation inside stores.
- **Self-Checkout & Contactless Payment**
Seamless checkout experiences using UPI, NFC, or facial recognition without any human interaction or queues.
- **Voice-Enabled Shopping Assistance**
Incorporation of voice assistants like Alexa or Google Assistant for hands-free operation and guidance while shopping.
- **Augmented Reality (AR) Navigation**
AR-based directions inside supermarkets to guide customers to product locations, offers, or sections.
- **Cloud-Based Billing & History Tracking**
Centralized cloud platforms to store purchase history, analyze user behavior, and offer dynamic discounts.
- **Integration with Smart Shelves**
Syncing trolleys with shelves that auto-detect product removal and restocking needs using sensors.
- **RFID-Enabled Theft Detection**
Enhanced security with real-time theft detection and automatic alerts triggered by unauthorized item removal.
- **Sustainability Monitoring**
Integration of tools to promote eco-friendly choices, such as alerts for plastic packaging or local products.
- **Autonomous Shopping Trolleys**
Development of self-driving shopping carts that follow customers or navigate independently within stores.

7.3. COMMERCIAL IMPLEMENTATION:

- **Retail Chain Integration**
Deploy in large supermarkets and grocery stores to reduce manpower, speed up checkout, and improve customer satisfaction.

- **Automated Billing Counters**
Replace or assist traditional billing counters by automatically transferring cart data for quick payment processing.
- **Customer Loyalty Programs**
Link the trolley system with customer accounts to track purchases, apply loyalty points, and offer personalized discounts.
- **Inventory Management**
Real-time data collection helps in managing stock levels, restocking alerts, and optimizing supply chains.
- **Data Analytics for Business**
Collect customer behavior and product preference data for business insights and targeted marketing.
- **Franchise-Based Solutions**
Offer the system as a modular product to retail franchises, enabling rapid deployment in multiple locations.
- **Rental Model for Small Stores**
Implement a rental/subscription model for small retailers who can't afford full ownership, making the system accessible to more businesses.
- **Self-Service Zones in Malls**
Install in mall outlets and food stores where customers can check out items independently using smart trolleys.
- **Integration with ERP Systems**
Synchronize with enterprise resource planning (ERP) software for seamless financial and logistical operations.
- **Branding & Advertisement**
Use LCD or touchscreens on trolleys to display ads or in-store promotions, creating new revenue channel.

CHAPTER-VIII

CONCLUSION

The Smart Shopping Trolley project demonstrates a practical and innovative approach to transforming the traditional retail shopping experience using embedded systems and automation. By integrating RFID technology, Arduino microcontrollers, GSM modules, and LCD interfaces, the system enables automatic item detection, real-time billing, and SMS-based notifications, significantly reducing the need for manual intervention at checkout counters. Through successful implementation and testing, the project has proven its ability to streamline the shopping process, enhance customer

convenience, and improve billing accuracy. Moreover, the modular design allows easy upgrades and integration with future technologies such as mobile apps, online payments, and cloud-based analytics. Despite a few limitations like network dependency and scalability challenges, the Smart Shopping Trolley presents a scalable and cost-effective solution for smart retail. With future enhancements such as IoT integration, AI-based personalization, and autonomous navigation, this system has the potential to redefine the way people shop in retail environments. Overall, this project lays a strong foundation for modern, efficient, and user-friendly shopping systems that can be adopted across supermarkets, malls, and other commercial outlets.